

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 1.2: Indices and Logarithms — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.2: Indices and Logarithms
Driving Question	DRIVING QUESTION: How can a Kericho tea farmer — or any Kenyan scientist, engineer, or business person — use the mathematics of indices and logarithms to make sense of, and calculate with, numbers that are too large or too small to handle by ordinary arithmetic?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Reciprocals and the Language of Indices: Launching the Tea Farm Phenomenon	Students observed a side-by-side comparison of long multiplication versus a four-figure logarithm table calculation for the Kericho tea farm total harvest (2^3 grams per plant $\times 2^{10}$ plants). They also completed a powers-of-2 table (2^0 to 2^{10}) by repeated doubling and extended it into negative exponents (2^{-1} , 2^{-2}) by observing the halving pattern.	Students learned that: (1) the reciprocal of a real number a is $1/a$, and $a \times (1/a) = 1$; (2) index notation expresses repeated multiplication compactly (e.g., $2^3 = 2 \times 2 \times 2 = 8$ grams per tea plant); (3) negative indices represent reciprocals — $a^{-n} = 1/a^n$; (4) increasing an exponent by 1 doubles the value (when base = 2), and decreasing it by 1 halves the value; (5) the anchoring phenomenon puzzle — how logarithm tables turn multiplication into addition — remains open and will be resolved across the unit.	Students explained, using the Kericho tea farm context, that: (1) the fraction of the estate's total production contributed by one single plant can be written as both $1/2^{10}$ and 2^{-10} — connecting reciprocals to negative index notation; (2) each additional tea plant exponent represents a doubling of harvest, which is why index notation is far more efficient than writing out large numbers; (3) their Draft Model 1 shows the conceptual links between reciprocals, negative indices, and the doubling pattern, with the logarithm mystery explicitly marked as 'still to be explained' — anchoring the inquiry arc for the remaining lessons.
Lesson 2 Deriving the Laws of Indices — Part 1: Product Rule, Quotient Rule, and Zero Index Law	Learners expanded index expressions (e.g., $2^3 \times 2^4$, $3^5 \div 3^2$, $5^2 \div 5^2$) into repeated-multiplication form, counted factors, cancelled common factors in quotient expressions, and recorded numerical results in structured discovery tables — all using examples drawn from Kericho tea-farm yields, Rift Valley land areas, and Nairobi/Kisumu population counts.	By expanding and counting factors systematically, learners derived three laws of indices: (1) $a^m \times a^n = a^{m+n}$ (product rule — exponents add when multiplying same-base powers); (2) $a^m \div a^n = a^{m-n}$ (quotient rule — exponents subtract when dividing same-base powers); (3) $a^0 = 1$ for any non-zero base a (zero index law — any base raised to zero equals 1, proved by the quotient rule when $m = n$). These laws	Learners explained their derived laws by applying them to Kenyan-context problems (Rift Valley plot areas, Kisumu fish quantities, mandazi recipes), co-constructing formal symbolic notation from group conjectures, annotating their Index Toolbox diagrams with all three laws and self-chosen worked examples, and posting questions on the class Driving Question Board — including the anticipatory question

		explain why changing the exponent on the Kericho farmer's tea-plant count by 1 doubles or halves the harvest.	of how the product rule connects to the farmer's logarithm addition trick.
Lesson 3 Deriving the Laws of Indices — Part 2: Power, Negative, and Fractional Indices	Learners observed (through their own group derivations) that: (1) raising a power to another power multiplies the exponents — $(2^3)^2 = 2^6 = 64$, confirmed by writing out all six factors of 2; (2) applying the quotient rule to a case where the denominator exponent exceeds the numerator (e.g. $2^3 \div 2^5 = 2^{(-2)} = 1/4$) reveals that negative indices produce reciprocal fractions; (3) using the power rule in reverse, $2^{(1/2)}$ must be the number whose square is 2, i.e. $\sqrt{2} \approx 1.414$ — meaning a 'half harvest cycle' yields the geometric mean between 2^0 and 2^1 .	Learners learned and can now state: (1) Power Rule — $(a^m)^n = a^{(mn)}$; (2) Negative Index Rule — $a^{(-n)} = 1/a^n$, so a negative exponent signals a reciprocal, NOT a negative quantity; (3) Fractional Index Rule — $a^{(1/n)} = \sqrt[n]{a}$ and $a^{(m/n)} = \sqrt[n]{(a^m)} = (\sqrt[n]{a})^m$. Together with the product rule, quotient rule, and zero index rule from Lesson 2, learners now hold the complete set of six index laws, all derived from the single foundational idea of repeated multiplication.	Learners explained: (1) why the Kericho tea farm manager's harvest figures grow exponentially — each additional power of 2 doubles production, and the power rule explains why scaling an already-scaled estate multiplies the exponent; (2) why $2^{(-3)}$ container-loads does not mean 'negative tea' but means the farm is $1/8$ of a reference container-load — a physically meaningful fraction relevant when comparing micro-plots to large estates; (3) why $2^{(1/2)}$ grams per 'half harvest cycle' is approximately 1.414 grams — the geometric midpoint of the doubling sequence — and how this connects roots to indices, foreshadowing logarithms as the tool for reversing any index operation.
Lesson 4 Introduction to Common Logarithms: Definition, Characteristic, Mantissa, and Reading Four-Figure Tables	Learners observed a pH data table featuring common Kenyan foods and liquids (lemon juice, uji, Rift Valley dairy milk, Lake Naivasha water) in which hydrogen ion concentrations were expressed as negative powers of 10. They noticed that the pH value of each substance equals the magnitude of the exponent of its $[H^+]$ concentration, and that a change of 1 pH unit corresponds to a tenfold change in concentration — mirroring the tea-farm phenomenon where changing an exponent by 1 doubles the harvest.	Learners learned the formal definition of the common logarithm: $\log_{10} x = n$ if and only if $10^n = x$. They learned to decompose any common logarithm into its characteristic (the integer part, equal to one less than the number of digits before the decimal point for numbers ≥ 1) and its mantissa (the non-negative decimal part, read from a four-figure table). They practised reading the mantissa from a physical four-figure logarithm table for eight numbers including 3,142 ($\pi \times 1,000$, connected to circular plantation geometry).	Learners explained that the logarithm is simply the exponent rewritten on the left side of the equation — the same exponent they have been working with since Lesson 1 in index notation. They articulated why the characteristic of $\log_{10} 3,500$ is 3 (because $10^3 \leq 3,500 < 10^4$) and connected the negative-characteristic logarithm $\log_{10} 0.01 = -2$ to the pH phenomenon (lemon juice $[H^+] = 10^{-2}$ mol/L). They began populating the LOGARITHM SIDE of their cumulative unit model and left the 'add logs $\rightarrow$ antilog $\rightarrow$ product' boxes blank as a bridge to Lesson 5.
Lesson 5 Using Mathematical Tables and Calculators for Logarithms and Antilogarithms	Students observed a teacher-modelled, narrated demonstration of the full four-step antilogarithm table procedure, applied to the M-Pesa transaction volume of 17,000,000 ($\log_{10} = 7.2304$). They also observed that pressing the 10^x key on a scientific calculator gives the same result as the antilog table, confirming that both tools are performing the same mathematical operation — raising 10 to a power.	The antilogarithm of x is 10^x ; finding it requires separating the characteristic from the mantissa, looking up the mantissa in the antilog table (row = first two decimal digits, column = third decimal digit, mean-difference = fourth decimal digit), and using the characteristic to place the decimal point in the result. Together with the log table procedure from Lesson 4, students can now make the full two-way journey between any number and its common logarithm.	The anchoring phenomenon — how the Kericho farmer's colleague multiplied two large harvest numbers in under two minutes using only a table — can now be traced step by step: (1) find log of each number using the log table; (2) add the two logarithms (a step students have used but not yet proved); (3) find the antilog of the sum using the antilog table to recover the product. Today's lesson provides step 3, completing the procedure. The proof that

			step 2 is valid ($\log(ab) = \log a + \log b$) is the open question that motivates Lesson 6.
Lesson 6 Applying Logarithms to Multiplication and Division	Learners observed two side-by-side worked examples — long multiplication of large tea-harvest numbers versus the same calculation performed using four-figure logarithm tables — and noted that the logarithm method replaced a multiplication step with an addition step, producing the same result in a fraction of the time.	Learners derived the product law $\log(ab) = \log a + \log b$ and the quotient law $\log(a/b) = \log a - \log b$ directly from the index law $10^a \times 10^b = 10^{(a+b)}$, and applied both laws using four-figure log and antilog tables to solve real-life Kenyan problems involving bulk maize pricing at Wakulima Market, fertiliser distribution in the Rift Valley, and annual tea revenue for a Kericho KTDA estate.	The reason multiplying two numbers is equivalent to adding their logarithms is that logarithms are exponents of powers of 10, and multiplying powers of 10 means adding their exponents — so $\log(M \times N) = \log M + \log N$ follows directly from the index law. This is exactly the mathematical principle that allowed the fellow Kericho farmer to complete the tea-harvest total production calculation in under two minutes using only a log table, converting an unwieldy multi-digit multiplication into a simple decimal addition.
Lesson 7 Applying Logarithms to Powers and Roots — Unlocking the Exponent and the Square Plot	Learners observed that computing 1.045^{12} by ordinary repeated multiplication is impractical and error-prone, while the same calculation — framed as $12 \times \log 1.045$ followed by an antilog lookup — takes under three minutes using four-figure tables. They also observed that finding $\sqrt{8,464}$ reduces to halving the logarithm of 8,464 and looking up the antilog, yielding a side length of approximately 92.0 m for a Nakuru County square plot.	The power law of logarithms, $\log(a^n) = n \cdot \log a$, follows directly from applying the product law n times — mirroring the way a^n means a multiplied by itself n times. The root law, $\log(\sqrt[n]{a}) = (1/n) \cdot \log a$, is an immediate corollary using the fractional-index rule $a^{(1/n)} = \sqrt[n]{a}$ from Lesson 3. Together, these two laws mean that any power or root calculation can be converted into a single multiplication or division of a logarithm, performed entirely with the four-figure tables introduced in Lessons 4 and 5.	The Kericho farm manager's logarithm-table method is now fully explained for all four operations. The 'magic' of adding small decimals to get large-number products (Lessons 5–6) extends to powers and roots: by scaling a small decimal by n (or by $1/n$), the table converts an exponential calculation — such as compound interest over 12 years — into arithmetic a secondary-school student can do in minutes. This completes the learners' understanding of why the fellow farmer at the technical training could solve the container-harvest problem 'in under two minutes without a calculator.'
Lesson 8 Consolidation, Combined Operations, and Model Revision: Answering the Driving Question	Learners observed two combined-operations logarithm problems representing a Kericho tea estate's blended-yield export calculation — a single expression involving multiplication, a cube power, division, and a square root, solved step-by-step using a four-figure mathematical table. They observed that every step could be reduced to addition or subtraction of logarithm values read from the same table, and that the final answer was recovered by a single antilogarithm look-up.	Learners learned that all four logarithm laws — Product, Quotient, Power, and Root — can be applied in sequence within a single multi-step expression; that the structured log-table format (Number Log Operation Result) organises this process reliably; that bar numbers arising from small decimals must be handled carefully when multiplied by a power; and that the entire system of indices and logarithms forms a coherent, interconnected toolkit for computing with very large or very small numbers efficiently.	Learners explained, in writing, how the mathematics of indices and logarithms allows a Kericho tea farmer — or any Kenyan scientist, engineer, or business person — to make sense of and calculate with numbers that are too large or too small for ordinary arithmetic: index notation compresses repeated multiplication into a compact exponential form; logarithms exploit the inverse relationship between exponentiation and its inverse to convert multiplication into addition; and the four-figure table provides a practical, calculator-free implementation of this conversion, reducing a long multiplication to a two-

			minute addition — exactly as demonstrated in the anchoring phenomenon.
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